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Paleomagnetism of the Acambay graben, central Trans-Mexican Volcanic Belt

Ana María Soler-Arechalde *, J. Urrutia-Fucugauchi

Laboratorio de Paleomagnetismo y Geofisica Nuclear, Instituto de Geofisica, Universidad Nacional Autonoma de Mexico, Del. Coyoacán 04510 D.F., Mexico

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Abstract

Paleomagnetic results for Miocene to Quaternary volcanic units of the Acambay graben are used to investigate the Neogene tectonic activity within the central sector of the Trans-Mexican Volcanic Belt (TMVB). Characteristic magnetization directions were obtained for 22 sites, with an overall mean direction (Dec = 176.7°, Inc = −34.3°, k = 28, $\alpha_{95} = 6°$) that is concordant with the expected direction calculated from North American reference data. Examination of site-mean directions shows an apparent geographic pattern related to the intersection of the Queretaro-Taxco fault zone with the E–W border fault system of the graben. This pattern is characterized by easterly and westerly declinations arranged into two broad groups. The mean direction for group A (easterly) is (n = 9): Dec = 189.9°, Inc = −28.5°, k = 53 and $\alpha_{95} = 4.8$. The mean direction for group B (westerly) is (n = 13): Dec = 167.6°, Inc = −36.2°, k = 38 and $\alpha_{95} = 4.3°$. Group A sites cover two areas, one inside the graben and the other immediately to the south including the Pastores fault. Group B sites are distributed over broad NW and SE areas across the graben, which include the Amealco caldera and Epitacio Huerta and Acambay–Tixmadeje faults and also the Venta de Bravo fault. Eighteen units show reverse polarities, and four units show normal polarity. Absence of normal polarity Quaternary units suggests a pre-Brunhes age for units sampled. K–Ar dating in the Amealco caldera gives a range between 5.7 and 2.2 Ma. A comparison of group A and B mean directions with reference directions calculated from the North American 10 Ma pole and the geographic pole gives rotation parameters between $9.1 \pm 5.3°$ and $9.9 \pm 4.6°$ and between $-12.4 \pm 5.4°$ and $-12.4 \pm 4.3°$, respectively. These may be interpreted in terms of vertical-axis rotations associated with regional left-lateral shear in a normal/strike-slip fault environment, which has characterized the Acambay graben during the Neogene. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: extensional tectonics; Neogene; paleomagnetism; Trans-Mexican Volcanic Belt

1. Introduction

Volcanic activity in Mexico is mainly concentrated along an east–west-elongated province, referred to as the Trans-Mexican Volcanic Belt (TMVB), which crosses the country from the Pacific Ocean to the Gulf of Mexico. The TMVB is associated with active subduction of the Cocos plate beneath the continental margin of southern Mexico along the Middle America trench. Volcanic products are of dominant calc-alkaline composition typical of the circum-Pacific continental and island arc provinces (e.g. Molnar and Sykes, 1969; Urrutia-Fucugauchi and Del Castillo, 1977). The

* Corresponding author. Tel.: +52-5622-4113.
E-mail address: anesoler@tonatiuh.igeofcu.unam.mx (A.M. Soler-Arechalde)

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TMVB includes large stratovolcano complexes, extensive monogenetic cinder cone and shield volcano fields and silicic centers.

The apparent structural control on the location and characteristics of volcanic activity in the TMVB has been discussed in several studies (e.g. Mooser, 1972; Pasquaré et al., 1986, 1988; Urrutia-Fucugauchi and Böhnel, 1988; Johnson and Harrison, 1990). Tectonic activity in the TMVB has been related to a regional east–west sinistral shear active since Miocene times and to the development of a progressively dominant extensional tectonism (Urrutia-Fucugauchi, 1984; Pasquaré et al., 1986; Delgado-Granados, 1993). Paleomagnetic studies have documented discordant directions with consistently counterclockwise rotated declinations with respect to reference North American data in the eastern and central sectors of the TMVB (Urrutia-Fucugauchi, 1984; Urrutia-Fucugauchi and Böhnel, 1988). Discordant declinations have been interpreted in terms of counterclockwise vertical-axis rotations of crustal blocks within the magmatic arc, consistent with left-lateral regional shear and transtension. The tectonic evolution of the TMVB is, however, not understood, and because of the length (~1000 km) of this continental arc and the marked contrasts among the different sectors, broad generalizations for the entire arc remain highly speculative. The relative motion with respect to North America of the Rivera and Cocos plates and southern Mexico blocks has proved difficult to quantify, and uncertainties remain (e.g. DeMets and Stein, 1990; Johnson and Harrison, 1990; Bandy and Pardo, 1994). The proposition of significant trench-parallel transport of blocks and slivers along the margin and along the trace of the TMVB is important for the plate kinematic evolution of the region.

A critical area that has yet to be studied is that of the central portion of the arc. This sector displays strong evidence of intense neotectonic activity along major east–west faults (Fig. 1). Taking this into consideration, attention is concen-

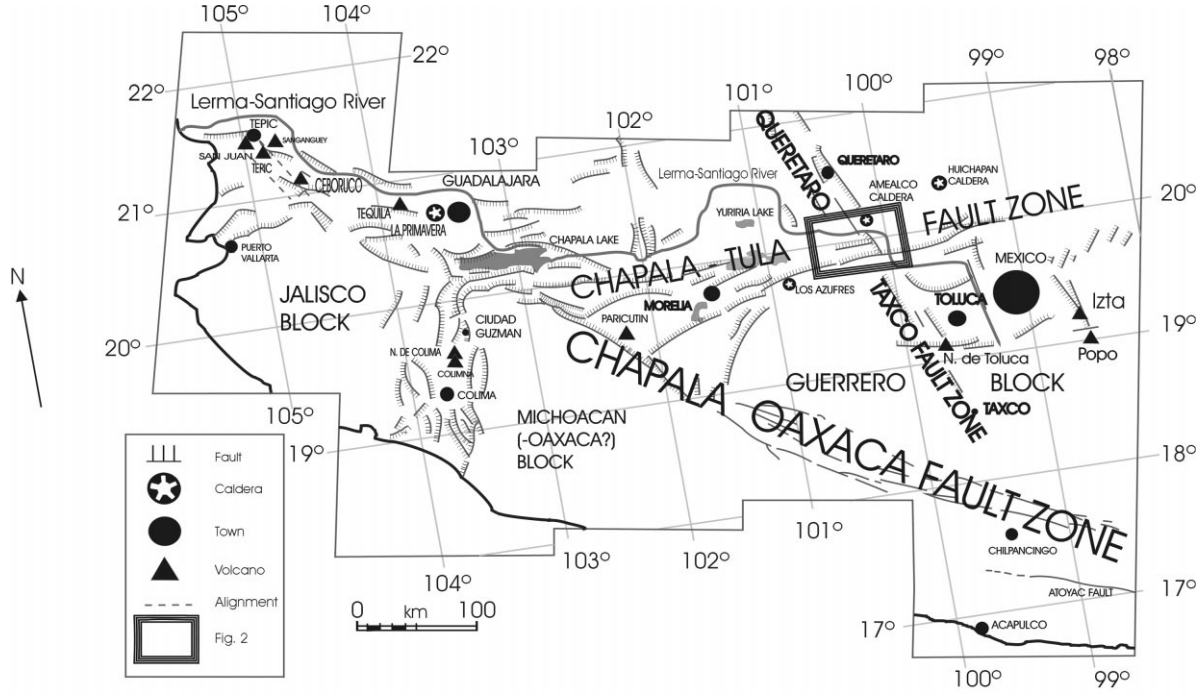


Fig. 1. Schematic tectonic map for west-central Mexico (adapted from Johnson and Harrison, 1990), showing location of the Acambay graben (small rectangle). The study area in the Acambay Graben lies at the intersection of two major fault zones: the E–W Chapala-Tula zone and the NW–SE Queretaro–Toluca zone, within the central sector of the Trans-Mexican Volcanic Belt (TMVB).

trated along the Chapala–Tula fault zone and along the Chapala–Oaxaca fault zone (roughly parallel to the trench), which may take most of the transport and deformation associated with plate interactions (DeMets and Stein, 1990; Johnson and Harrison, 1990). In this paper, we summarize the results of a paleomagnetic study of volcanic units in the Acambay graben, located at the easternmost sector of the Chapala–Tula fault zone (Fig. 1). The Acambay graben region is characterized by an intense neotectonic, volcanic and historic shallow crustal seismic activity (e.g. Urbina and Camacho, 1913; Astiz, 1980; Suter et al., 1992).

2. Acambay graben and paleomagnetic sampling

The Acambay graben is located in the central sector of the TMVB (Figs. 1 and 2) and is characterized by long, generally E–W depressions and fault scarps. It forms part of the regional fault system, referred to as Chapala–Tula by Johnson and Harrison (1990). Historic shallow crustal seismicity characterizes some of the major faults in the region; with the 19 November 1912, $M_s = 7.0$, Acambay earthquake (Urbina and Camacho, 1913; Suter et al., 1992) and the 22 February 1979, $m_b = 5.3$, Maravatio earthquake (Astiz, 1980).

The region studied extends from Maravatio in the west into the Acambay graben in the east (Fig. 2) and includes a major NNW–SSE 'discontinuity zone' that extends from the Amealco caldera to the Perales fault. This zone is clearly marked in the gravity anomaly field; the regional pattern of Bouguer anomalies change from a broad E–W trend to a NNW–SSE orientation coinciding with the Amealco–Perales deformation zone (Molina-Garza and Urrutia-Fucugauchi, 1993; Urrutia-Fucugauchi et al., 1993) (Queretaro-Taxco fault system). The Lerma river follows this change in orientation showing a structural control in the region (Fig. 1).

For the paleomagnetic study, we collected oriented samples from 22 sites distributed in the Acambay region (Soler-Arechalde, 1990, 1997). Samples were drilled in the field with a portable petrol-powered drill and oriented in situ with a Brunton compass. Sites were located along fault scarps, and at exposures and road-cuts into lava flows and pyroclastic units (Fig. 2). Volcanic units in the area are essentially flat-lying, with some units sampled dipping some 4°. Volcanic units in the region range in age from the Miocene to

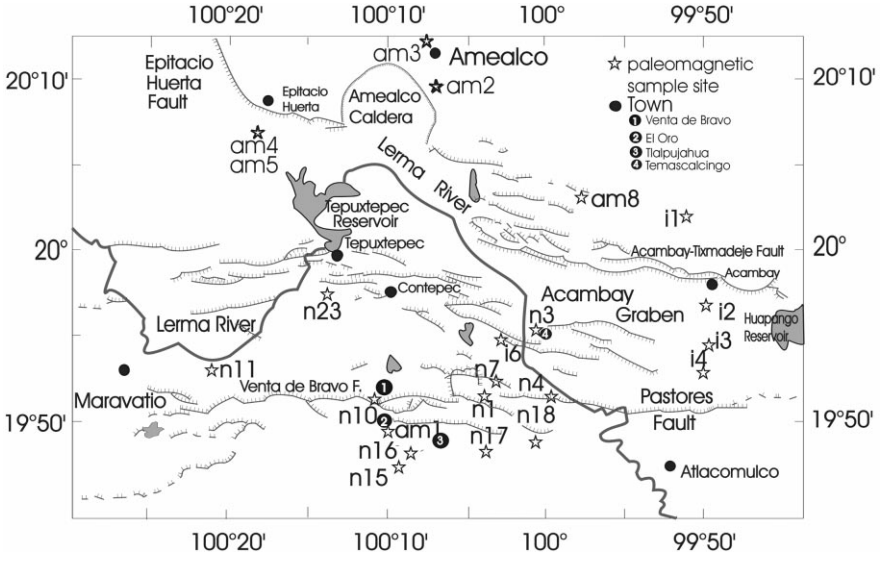


Fig. 2. Schematic simplified tectonic map of the Acambay graben (modified from Suter et al., 1992) with location of paleomagnetic sampling sites (stars).

Table 1
Summary of palaeomagnetic data for the Acambay graben, central Mexico

Site	n	Dec	Inc	r	k	α_9	R	ΔR	F	ΔF	Age
n1	8	190.1	−11.5	7.9740	269	2.7	10.1	2.8	−24.3	2.7	Pl
n3	4	158.9	−29.6	3.9828	174	5.6	−21.1	6.4	−6.3	5.6	Mc
n4	5	181.9	−45.0	4.8985	39	9.9	3.2	14.0	11.1	9.9	Mc
n7	5	161.8	−45.2	4.8854	35	10.0	−17.4	14.2	12.1	10.0	Q
n10	7	155.1	−43.8	6.7438	23	10.2	−24.9	14.1	7.9	10.2	Q
n11	7	147.4	−37.9	6.8431	38	7.9	−32.6	10.0	2.0	7.9	Q
n15	6	353.3	24.2	5.9369	79	6.1	−6.7	6.7	11.6	6.1	Mc
n16	6	358.2	35.8	5.9740	192	3.8	−1.8	4.7	−0.03	3.8	Mc
n17	8	183.5	−21.6	7.9529	149	3.7	3.5	4.0	−14.2	3.7	Mc
n18	9	193.8	−34.3	8.6646	24	8.6	13.8	10.4	−1.5	8.6	Mc
n23	6	173.1	−15.3	5.8435	32	9.6	−6.1	10.0	−17.7	9.6	Q
i1	6	164.7	−48.6	5.8688	38	8.8	−15.3	13.3	12.4	8.8	Pl
i2	6	167.9	−46.2	5.9689	161	4.2	−12.1	6.1	11.5	4.2	Mc
i3	6	192.7	−20.2	5.9431	88	5.8	12.7	6.2	−15.7	5.8	Pl
i4	4	192.3	−32.1	3.9788	141	6.2	12.3	7.3	−3.7	6.2	Pl
i6	8	183.5	−39.1	7.9867	528	1.9	3.5	2.5	3.2	1.9	Q
am1	12	190.9	−35.5	11.9268	150	2.9	10.9	3.6	−0.3	2.9	Mc
am2	9	179.0	−45.4	8.9799	397	2.1	−1.0	3.0	9.1	2.1	Mc
am3	8	351.4	29.3	7.9485	136	3.8	−8.6	4.4	7.1	3.8	Mc
am4	9	351.2	21.0	8.9621	211	2.9	−8.8	3.1	15.2	2.9	Mc
am5	4	191.9	−33.4	3.9656	87	7.9	11.9	9.5	−2.8	7.9	Mc
am8	11	172.8	−43.6	10.9657	289	2.2	−7.2	3.0	7.5	2.2	Mc
Mn1	22	176.7	−34.3	21.2487	28	6.0	−3.3	7.3	−1.7	6.0	

n: number of samples; Dec: declination; Inc: inclination; r: vector resultant; k, concentration parameter; α_{95}, 95% cone of confidence; $R \pm \Delta R(^*)$: rotation parameter and statistical uncertainty; $F \pm \Delta F(^*)$: flattening parameter and statistical uncertainty; Mn1: mean of all sites; (*): with respect to the geographic pole

Recent (Table 1) (Silva-Mora, 1979; Sanchez-Rubio, 1984; Aguirre-Díaz, 1993; Aguirre-Díaz et al., in press).

3. Paleomagnetic results

The direction and intensity of natural remanent magnetization (NRM) were measured using a Molspin fluxgate magnetometer. The vectorial composition and magnetic stability of NRM of all samples were investigated by detailed stepwise alternating field (AF) demagnetization using a Schonstedt GSD-5 demagnetizer. Two examples of behavior during demagnetization are illustrated in the ortogonal vector plots of Fig. 3. Most samples showed well-defined characteristic components of high directional stability, which likely correspond to thermoremanent magnetizations (TRM). Magnetic carriers correspond to iron-rich titanomagnetites. Experiments of acquisition of isothermal remanent magnetization (IRM) were completed in some selected samples from every site, by using a laboratory-made pulse magnetizer. Additional magnetic hysteresis experiments using small samples prepared from the cores were carried out using a MicroMag system (Figs. 4 and 5). As in the demagnetization curves (Fig. 3) and the IRM experiments, the majority of hysteresis loops show coercitivities corresponding to titanomagnetites, and only in a few cases were high coercitivity minerals (hematite?) observed. Results of the hysteresis measurements are summarized in a Day diagram (Day et al., 1977) showing the domain state (Fig. 6). Samples studied with the MicroMag fall within the pseudo-single domain (PSD) field with Hcr/Hc ratios between 1.5 and 2.8 and Mr/Ms ratios between 0.1 and 0.5 (Fig. 6). No samples are observed in the single domain (SD) or multi-domain (MD) fields. None of the small samples

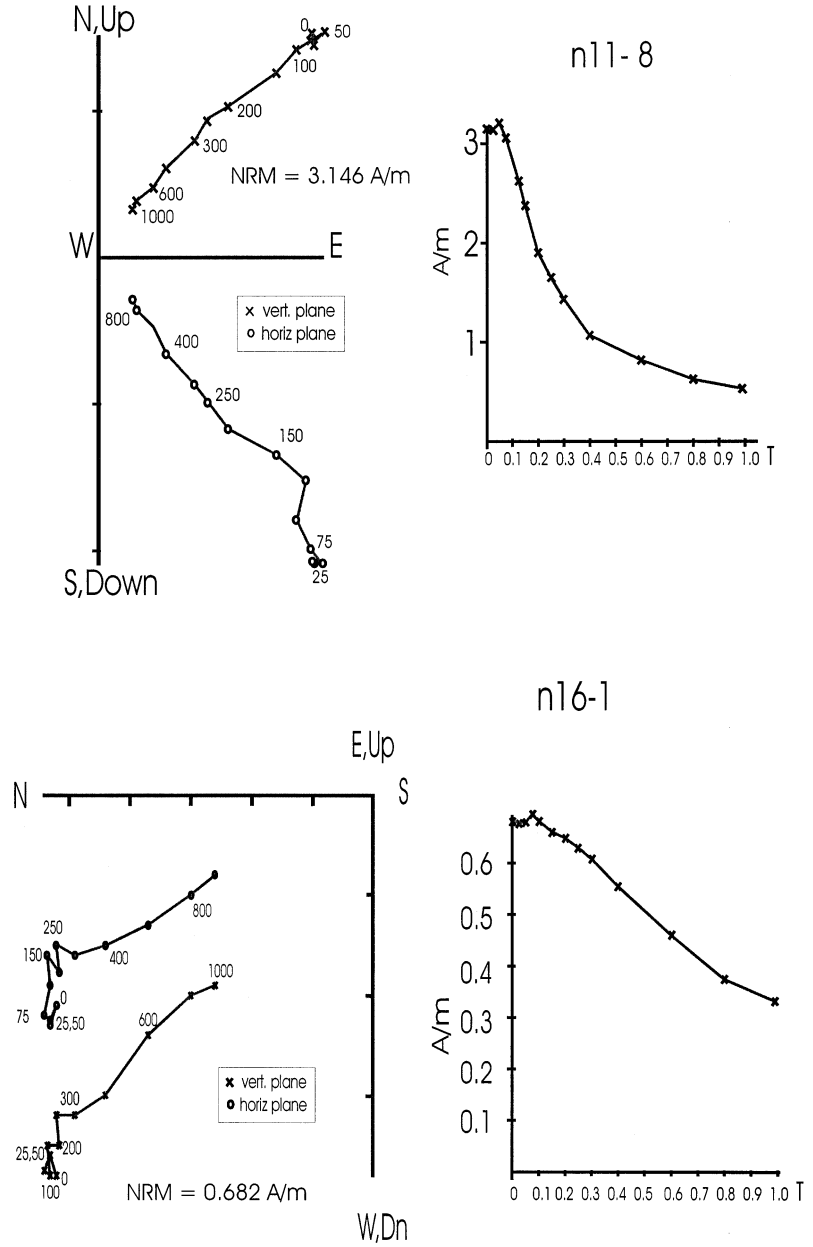


Fig. 3. Examples of Zijderveld vector plots and demagnetization curves of the samples n11-8 and n16-1, showing different behaviors obtained by step-wise alternating field demagnetization. Intensity values are normalized.

analyzed with the MicroMag showed the presence of high coercivity minerals.

Vector subtraction, end-point and principal component analysis (Zijderveld, 1967; Dunlop, 1979; Kirschvink, 1980) were used for calculating characteristic directions for each sample. Site mean directions and corresponding pole positions were calculated from the characteristic sample directions. Results and the corresponding Fisher statistical parameters (Fisher, 1953; Irving, 1964) are

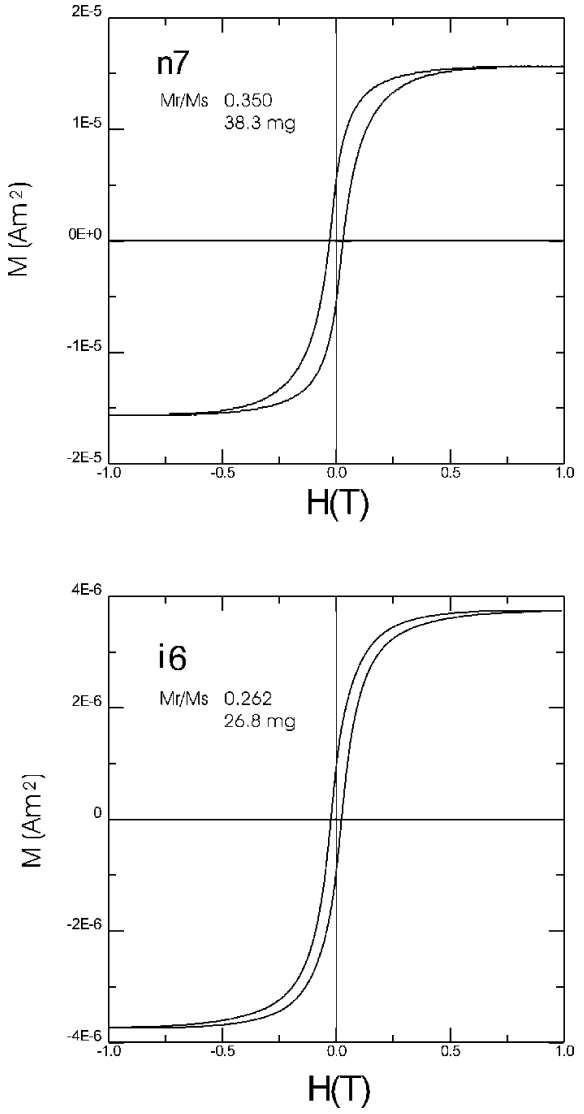


Fig. 4. Examples of hysteresis loops for two samples from sites n7 and i6 obtained using the MicroMag.

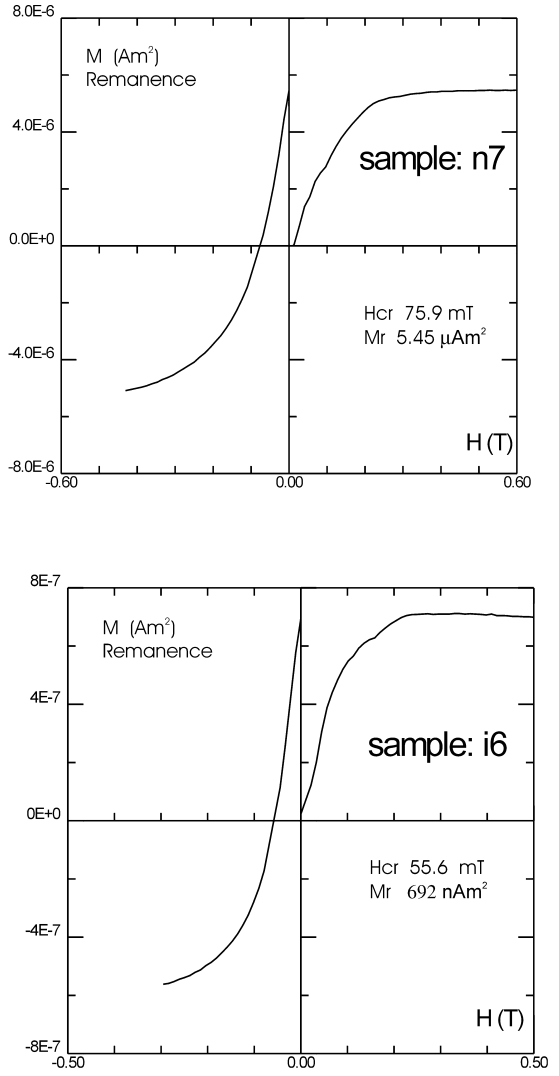


Fig. 5. Examples of isothermal remanent magnetization (IRM) acquisition diagrams and back-field demagnetization curves for two samples from sites n7 and i6.

included in Table 1. Site mean directions are illustrated in Fig. 7. Most sites show a reverse polarity, and only four sites show normal polarity.

The mean direction for the 22 sites is concordant with the direction estimated from the Neogene North America reference direction. Examination of site-mean directions shows the occurrence of anomalous declination values with respect to the expected Neogene direction for the region. The pole position used as the Neogene reference is the 10 Ma mean pole of Irving and Irving (1982) and the dipolar direction. To quantify the angular divergence, the rotation (R) and the flattening (F) parameters of Beck (1980) were calculated, together with their statistical uncertainties, ΔR and ΔF (Demarest, 1983). The R values (Table 1) vary from $-32.6^\circ \pm 10^\circ$ to $13.8^\circ \pm 10.4^\circ$. These anomalies fall into four spatial clusters, two with $R>0$ and two with $R<0$ (Fig. 8). The Neogene pole

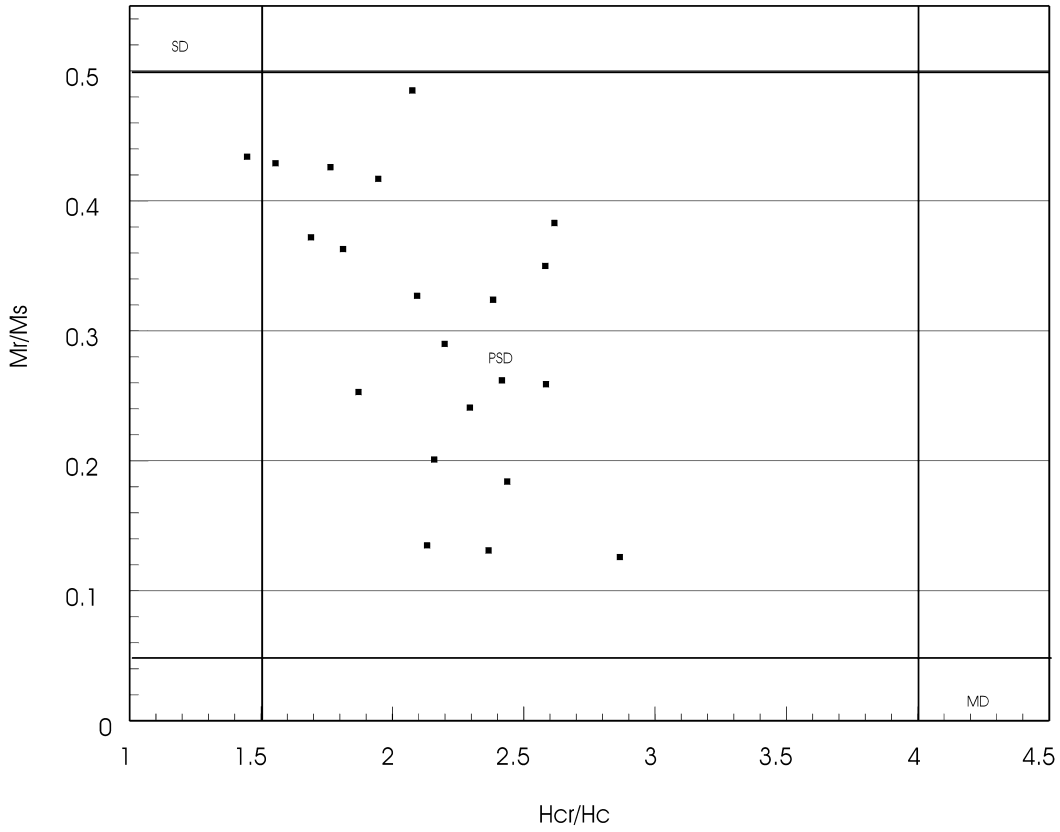


Fig. 6. Summary diagram for hysteresis parameters, showing the dominant domain states of samples (Day et al., 1977).

and geographic pole are used for calculations in Table 2.

4. Discussion

The Acambay graben region forms part of the larger E–W-oriented Chapala–Tula fault zone (Fig. 1). Analysis of Landsat Thematic Mapper (TM) images (Johnson and Harrison, 1990) and detailed field mapping (Suter et al., 1992; Ramírez-Herrera, 1998) show that the Acambay graben is characterized by numerous predominantly E–W faults, which define the major structure as well as smaller features (Fig. 2). The southern boundary is formed by the Venta de Bravo and Pastores faults, which are characterized by long continuous escarpments with up to 300 m of topographic relief. The separation between the Venta de Bravo and Pastores faults is the intersection with the NNW–SSE Amealco–Perales fault zone (part of the Queretaro–Taxco fault system) (Fig. 1) and is characterized by an en echelon fault pattern and small extensional jogs or pull apart basins. The northern boundary is marked by the Acambay-Tixmadejé fault system that exhibits fault scarps of up to 400–500 m of topographic relief. In the northwestern sector of the graben, the Epitacio Huerta fault affects the southern rim of the Amealco caldera. The fault changes direction near the village of Epitacio Huerta from NNW–SSE, coinciding with the direction of the older Queretaro–Taxco fault system, to E–W coinciding with the direction of the Chapala–Tula fault system (Fig. 2). The discontinuity between the Epitacio Huerta and the Acambay–Tixmadejé fault systems has been interpreted as a right stepover (Suter et al., 1992). North of Venta de Bravo fault are

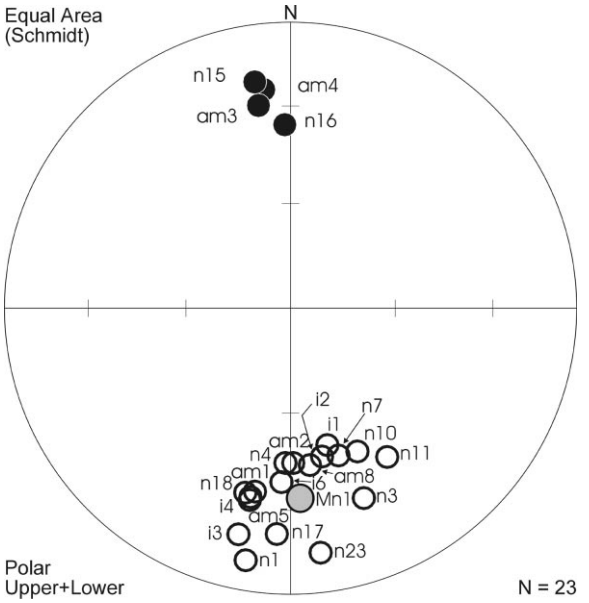


Fig. 7. Summary of site mean paleomagnetic directions (see Table 1). Open symbols correspond to negative inclinations, and full symbols correspond to positive inclinations. Note the predominance of reverse polarity, with only four sites of normal polarity. The large circle represents the overall mean direction Mn1 of Table 1.

the Tepuxtepec faults and the Temascalcingo faults (Ramírez-Herrera, 1998), a series of minor faults inside the graben. These faults show predominantly normal displacements, with a minor left-lateral strike-slip component like in the Pastores and Venta de Bravo faults (Astiz, 1980; Suter et al., 1992).

Examination of the rotation parameters in terms of spatial distribution within the graben shows that sites with similar sign of R values can be separated into four broad areas (Fig. 8), although in some cases, a gradual variation from one site to the next can be observed. Two areas with $R>0$ referred to as group A (with nine sites) and two areas with $R<0$ referred to as group B (with 13 sites). Group A has a mean direction ($n=9$) Dec = 189.9°, Inc = −28.5°, $k=63$ and $\alpha_{95}=5.7°$, and the corresponding pole is 80.6°N and 11.3°E. Group B has a mean direction ($n=13$), Dec = 167.6°, Inc = −36.2°, $k=38$ and $\alpha_{95}=5.4°$, and the corresponding pole is 78.4°N and 172.5°E.

A subdivision of group A covers an area inside and outside the graben that includes a section of the Pastores fault, some minor faults near Tlapujahua and El Oro towns and other minor faults near Temascalcingo (dark discontinuous line, Fig. 8); here, the Lerma River basin (dark continuous line, Fig. 1) follows the alignment direction of the Queretaro–Taxco fault system. The am1, n17 and n18 are sites of the same unit [the Amealco ignimbrite of Miocene age (Aguirre-Díaz, 1993)] and have a reverse polarity and similar declination values. The other sites of group A are: n4, a Pliocene dacite; n1, a Pliocene andesite; and i6, a Quaternary rhyolite (Silva-Mora, 1979). Their declination anomalies could reflect local effects, because they are located over fault scarps. The second subdivision of group A is located to the east and inside the graben structure. Two sites i3 and i4 are Quaternary basalts.

Group B is subdivided in two subgroups (light lines in Fig. 8). A major subdivision covers the middle and southwestern part of the area and includes the Venta de Bravo fault. Irrespective of their polarity and age (n23, n11, n10 and n7 reverse polarity and Quaternary age and n15 and n16, normal polarity and Miocene age) all sites have negative R parameters (i.e. counterclockwise rotations). The second subdivision is located in the middle and northeastern sectors, including the Acambay–Tixmadejé and Epitacio Huerta faults. The Lerma river, aligned with the NNW–SSE Queretaro–Taxco fault system, crosses the area. The majority of the sites are from the Miocene Amealco ignimbrite (sites am4, am3, am2, am8 and i1). The other sites of this group n3 and i2 are also of Miocene age (Silva-Mora, 1979; Sanchez-Rubio, 1984; Aguirre-Díaz et al., in press). All sites, except am3, of this subgroup present reverse polarities.

Declination anomalies could be interpreted in terms of apparent tectonic rotations, incompletely averaged paleosecular variation effects or vertical axis rotations (MacDonald, 1980). The volcanic units sampled are essentially horizontal, some dipping just some 4°. Therefore, no structural corrections have been applied, and no apparent rotations due to incorrect corrections can arise (MacDonald, 1980). Assuming that paleosecular variation effects have been averaged out and that no structural

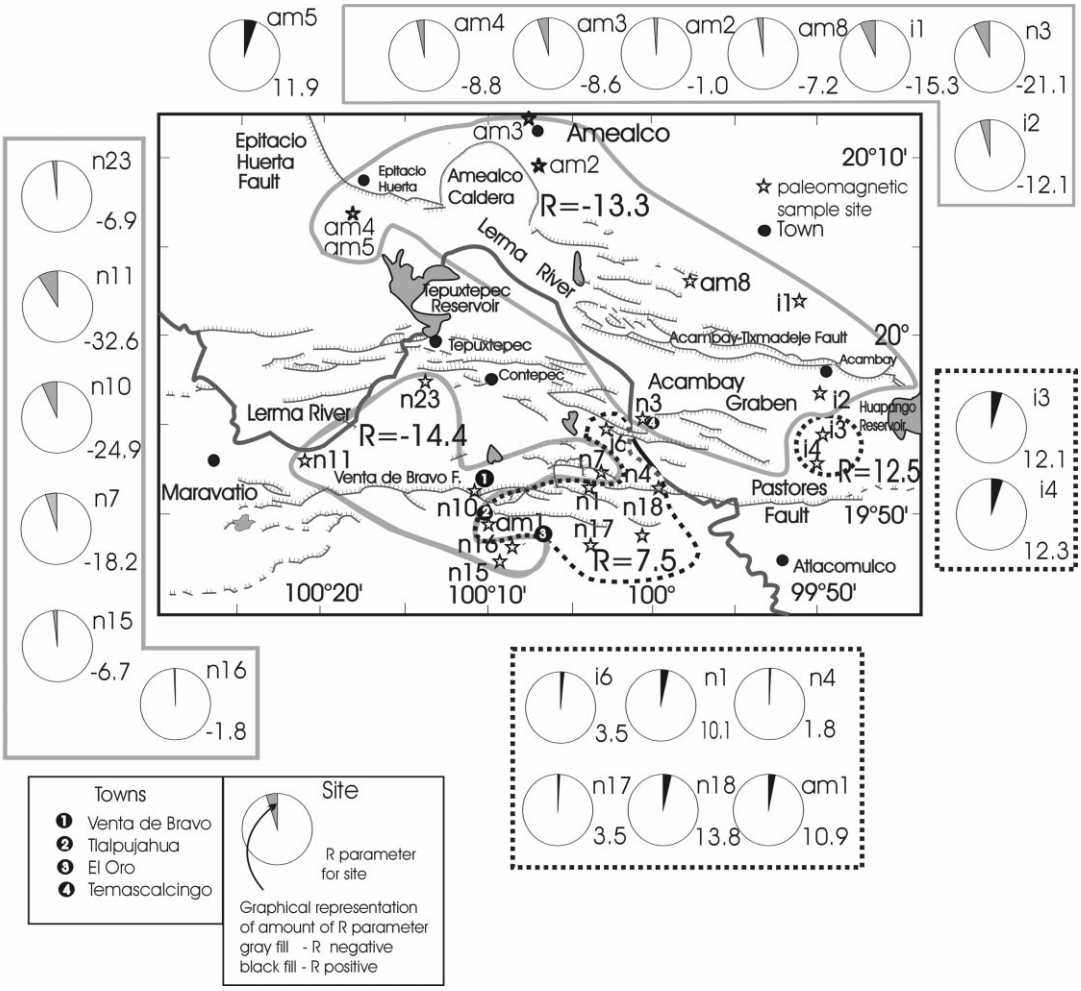


Fig. 8. Schematic simplified tectonic map of the Acambay graben (modified from Suter et al., 1992) with location of paleomagnetic sampling sites (stars) and the four tectonic domains proposed with their R parameter. The pie diagrams provide a visual summary of the sense and amount of discordance of the paleomagnetic data for each site. Black fill is for $Rs>0$ and gray fill is for $Rs<0$. Results are shown in terms of site mean declination and α_{95}. Most paleomagnetic declinations are predominantly of a reverse polarity and have been converted to normal polarity for visual display only to clarify comparison of the rotation angles. The complete paleomagnetic data set is given in Table 1. Lerma River: continuous darker line. Group A sites are enclosed by the dark discontinuous curve. Group B sites are enclosed by the continuous light curve.

correction is needed, the clustering of site mean directions within given geographic areas may indicate that the Acambay region is constituted by rotated domains, possibly in response to the regional left-lateral stress regime. Local differential rotations of smaller domains within the region may complicate the pattern.

To estimate absolute magnitude and timing of the tectonic rotations, detailed information on the age of the volcanic units is needed. Unfortunately, there is no detailed volcanic stratigraphy for the region, except for the Amealco ignimbrite (Aguirre-Díaz et al., in press). The predominance of reverse polarities indicates that volcanic units sampled (previously assigned to the Quaternary by previous stratigraphic studies) correspond to pre-Brunhes activity and may be assigned to the Pliocene or early Pleistocene. It is, however, diffi-

Table 2
Summary of palaeomagnetic directions for the Trans-Mexican Volcanic Belt

Locality	*n*	Lat (N)	Long (E)	α_{95}	$R \pm \Delta R$	$F \pm \Delta F$	Age	Ref.
1. E. Mexico volc.		80.1	184.1	8.1	-8 ± 11	-4 ± 8	M–Q	a
2. Basin Mexico	42	87.5	164.5	3.0	-2 ± 4	-3 ± 6	Q(B)	b
3. Basin Mexico	19	88.1	302.8	6.5	5 ± 7	-4 ± 6	eM–P	c
4. Basin Mexico	22	88.4	79.3	7.2	8 ± 8	6 ± 8	eO–M	c
5. Basin Mexico	271	80.7	154.3	5.3	-9 ± 6	4 ± 5	P–Q	d
6. Jantetelco-Tepexco	9	33.7	176.3	10.0	-56 ± 13	6 ± 13	M	e
7. NE Jalisco	7	68.1	181.1	(10.0)	-20 ± 14	1 ± 12	O–M	f
8. Guerrero volc.	6	54.8	164.5	(8.6)	-32 ± 11	11 ± 12	O–M	g
9. Balsas Fm.	14	54.1	183.4	(12.0)	-40 ± 14	18 ± 14	Pa–E	h
10. R. Santiago	3	81.2	128.1		-6	6	P	i
11. R. Santiago	4	79.1	180.6		11	-6	M	i
12. Amatlan volc.	8	81.9	197.2	(9.1)	-8		Q	j
13. E. Chapala L.	16	74.0	159.7	7.6	-17 ± 7	3 ± 6	M–Q	k
14. Volc. Mexico W	16	84.9	155.9	8.9	5	3	P–Q	l

n: number of samples; Lat: latitude; Long: longitude; α_{95}: 95% cone of confidence; $R \pm \Delta R$: rotation parameter and statistical uncertainty; $F \pm \Delta F$: flattening parameter and statistical uncertainty. Ages: Q: Quaternary: Q(B): Quaternary (Brunhes chron); P: Pliocene; M: Miocene; eM: early Miocene; O: Oligocene; eO: early Oligocene; E: Eocene; Pa: Paleocene. References (Ref.): a: Ruíz-Martínez et al. (in press); b: Herrero-Bervera et al. (1986); c: Mooser et al. (1974); d: Osete et al. (in press); e: Urrutia-Fucugauchi (1981); f: Urrutia-Fucugauchi and Pal (1977); g: Urrutia-Fucugauchi (1983); h: Urrutia-Fucugauchi (1980); i: Watkins et al. (1971); j: Nieto-Obregón et al. (1992); k: Rosas-Elguera and Urrutia-Fucugauchi (1992); l: Maillol and Bandy (1994).

cult to make a detailed correlation with the geomagnetic polarity time scale. The reverse polarities, irrespective of lithology, imply a revision of the age of volcanic activity in the region that is currently considered as Quaternary. The older age indicated by the paleomagnetic polarities corresponds better with two reported K–Ar dates for the Amealco ignimbrite (5 Ma) and Yondeje andesite (13 Ma), both with reverse polarities possibly corresponding to reverse chrons between anomalies 3 and 3A and 5A and 5B, respectively (Harland et al., 1990).

A simple model of vertical axis block rotations for the Acambay region to account for the clockwise and counterclockwise rotated declinations is examined using the tectonic patterns and geometric parameters of Fig. 9. The model is derived after models developed by Ron et al. (1984) and McKenzie and Jackson (1986, 1989). The premise in Ron's model is that in order to remain in contact with the boundary of the fault domain, the blocks must rotate and that the sense of rotation would be opposite to the sense of the fault slip. A left-lateral strike-slip direction is considered for the Venta de Bravo–Pastores and the Tepuxtepec fault, and a right-lateral component for the Epitacio Huerta and Acambay–Tixmadejé faults. The azimuth of the slip vector of the Cocos plate is $\Theta = 21.83^\circ$, the mean width (a) for the deformation area is around 45 km, and the eastern component (Wa) of the convergence vector of the Cocos plate is about 22.31 mm/yr. With these values, an R parameter of -14.1° is obtained. Using the McKenzie and Jackson (1986) model, R is $-12.4 \pm 5.4^\circ$. We note that Θ and the angle between the E–W faults and the Queretaro–Taxco fault is $\phi = 25^\circ$ (Fig. 9), which fulfills one of the requirements in the model of McKenzie and Jackson (1986). This emphasizes the role played by the oblique geometrical relationships between the fault systems Chapala–Tula and Queretaro–Taxco, as well as the convergence angle of the Cocos and North American plates.

Previous paleomagnetic studies in the TMVB have documented characteristic directions with declinations rotated counterclockwise with respect to expected directions (summarized in Table 3). The localities studied are, however, concentrated in the eastern sector or immediately south of the TMVB, and most of the magmatic arc has not

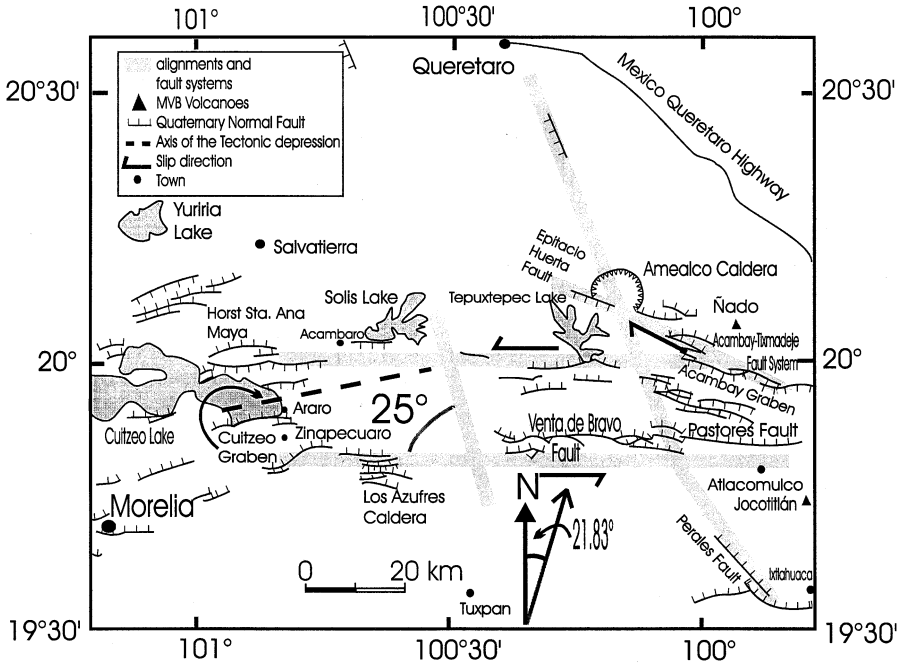


Fig. 9. Schematic diagram that shows the parameters used in the model of vertical axis block rotation: $\phi = 25^\circ$ (azimuth of the Queretaro-Taxco fault system), $\theta = 21.83^\circ$ (slip vector azimuth of Cocos plate), $Wa = 22.31$ mm/yr, $a = 45$ km (width of the deformation area) giving an R ('floating model') $= -14.2^\circ$/Ma. Note that ϕ and θ are concordant.

been investigated. Modeling of present-day plate kinematics in the Middle American region suggests the occurrence of a sinistral component of motion for southern Mexico in the order of 0–10 mm/yr (DeMets and Stein, 1990; Johnson and Harrison, 1990). This difference in relative motion between southern Mexico and the North American plate may be accommodated by southeastward and eastward displacement of southern Mexico (Fig. 1) along the trace of the magmatic arc (Chapala–Tula fault zone) and a trench-parallel fault (Chapala–Oaxaca fault zone). The trend of the Chapala–Tula zone is oblique to the trend of the Middle American trench and to the Cocos plate convergence direction. Volcanic activity in the arc is associated with E–W structures and seems to emphasize a dominant structural and deep crustal control for the magmas and lithospheric deformation.

Analyses of the stress field for central Mexico from analysis of Quaternary cinder cone alignments, inversion of striation measurements and

Table 3
Summary of palaeomagnetic rotation data for the Acambay graben, central Mexico[a]

Rotation	n	Dec	Inc	k	α_{95}	R NAm	ΔR	R Bruh	ΔR
Group A Clock	9[b]	189.9	−28.5	53	4.6	9.1	5.3	9.9	4.6
Group B C-Clock	13[c]	167.6	−36.2	38	4.3	−12.4	5.4	−12.4	4.3

[a] Clock: clockwise rotation; C-Clock: counterclockwise rotation; n: number of samples; Dec: declination; Inc: inclination; r: vector resultant; k: concentration parameter; α_{95}: 95% cone of confidence, α_{95}. Rotation parameter and statistical uncertainty $R \pm \Delta R$: with respect to the North American 10 Ma reference pole (Irving and Irving, 1982) and dipolar direction.

[b] Considered all sites with $R > 0$ in Table 1.

[c] Considered all sites with $R < 0$ in Table 1.

the focal mechanism of the 1979 Maravatio earthquake suggest that normal and left-lateral strike-slip faulting result from superposition of two stress sources (Suter et al., 1992). The regional left-lateral shear component is related to oblique convergence of the Cocos and North American plates. The amount of lateral displacement that can be accommodated by the large faults is relatively small, and it has been considered insufficient to account for the rigid block rotations. This situation has been found in other tectonically deformed areas such as in the San Andreas fault zone where tens of degrees of rotation have been documented. It has been pointed out that rigid rotation requires high strains at fault tips and formation of voids, and displacements approximately constant at the bounding boundaries of domains (e.g. Dibble, 1977; Peacock et al., 1998). We have recognized that similar problems are encountered to explain rigid block rotations in the Acambay graben, particularly the modest lateral displacement observed in the large faults. Alternative models that have been developed to account for rotations involve distributed deformation within blocks (e.g. Sonder et al., 1994). Distributed deformation may permit rotation without the formation of voids at boundaries between adjacent domains (Dibble, 1977). Voids formed at boundaries of rotated domains are present with the development of sedimentary basins (Luyendyk et al., 1980) or emplacement of intrusives (Tikoff and Teyssier, 1992). Peacock et al. (1998) have recently proposed that rotation may be accommodated by sets of small faults, which can take part of the deformation without the need for microscopic scale deformation of continuum models.

The paleomagnetic data showing the occurrence of rotations along the eastern Acambay sector of the regional E–W Chapala–Tula fault system are compatible with proposals for a major crustal weakness zone that provides favorable conditions for magmatic activity. The structural control on magmatism may explain the oblique geometric arrangement between the magmatic arc and the trench. In the past, several studies have proposed the presence of a major crustal discontinuity. Documentation of the characteristics of neotectonic deformation has wide implications for the interplay and relationships among the overriding plate, magmatism, tectonics and plate subduction in this continental magmatic arc (Pasquaré et al., 1988; DeMets and Stein, 1990; Johnson and Harrison, 1990). Western Mexico represents a transitional zone that has experienced repeated plate reorganizations in the past few million years (Mammerickx and Klitgord, 1982), where the dominant tectonics change from northward motion and clockwise rotations of margin terranes [including the modern Baja California and Californian terranes along the right-lateral San Andreas fault system; Beck (1980)] to the southeastward left-lateral movement, and associated counterclockwise rotations, that characterize the Middle and Central American regions (and Caribbean plate).

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